Duke University **November 18, 2021**
Math 105L: Laboratory Calculus and Functions I

Big Assignment 3

- This assignment covers work from Day 10-2 to Day 13-3.
- You have eleven days to complete the assignment. It is due November 29 at 8am.
- You may use the worksheets, labs, and your notes, as well as any other resources.
- However, you may not work with anyone else on this assignment – it must be entirely your own work.
- Throughout the assignment, show your work so that your reasoning is clear. Otherwise no credit will be given. Circle your answers. For example, copying graphs from graphing software is insufficient 'work'.
- Do not spend too much time on any one problem. Read them all through first and attack them in an order that allows you to make the most progress.
- You must hand in a scanned version of this assignment on Gradescope and no other pages. If you need to do rough work, do it elsewhere before writing your answers.
- The number of points on each question is shown next to the question number. There are 120 points available in total, plus four bonus points.
- You must sign the statement at the bottom of this page.

Honor Code:

I have adhered to the Duke Community Standard and all the rules above in completing this assignment.

Signature: _____

1. (34 points) Consider the function $f(x) = x \ln |x|$.

   (a) Why is $f(x)$ not defined at $x = 0$?

   (b) Show that $\frac{d}{dx} \ln |x| = \frac{1}{x}$ (Hint: write $|x|$ as a piecewise function!)

   (c) Show that $\lim_{x \to 0} x \ln |x| = 0$. Label where you use L'Hôpital's rule.

To make $f(x)$ continuous and defined on all real numbers, we now redefine it:

$$f(x) = \begin{cases} x \ln |x| & x \neq 0 \\ 0 & x = 0 \end{cases}$$

   (d) Show that $f'(x) = \ln |x| + 1$.

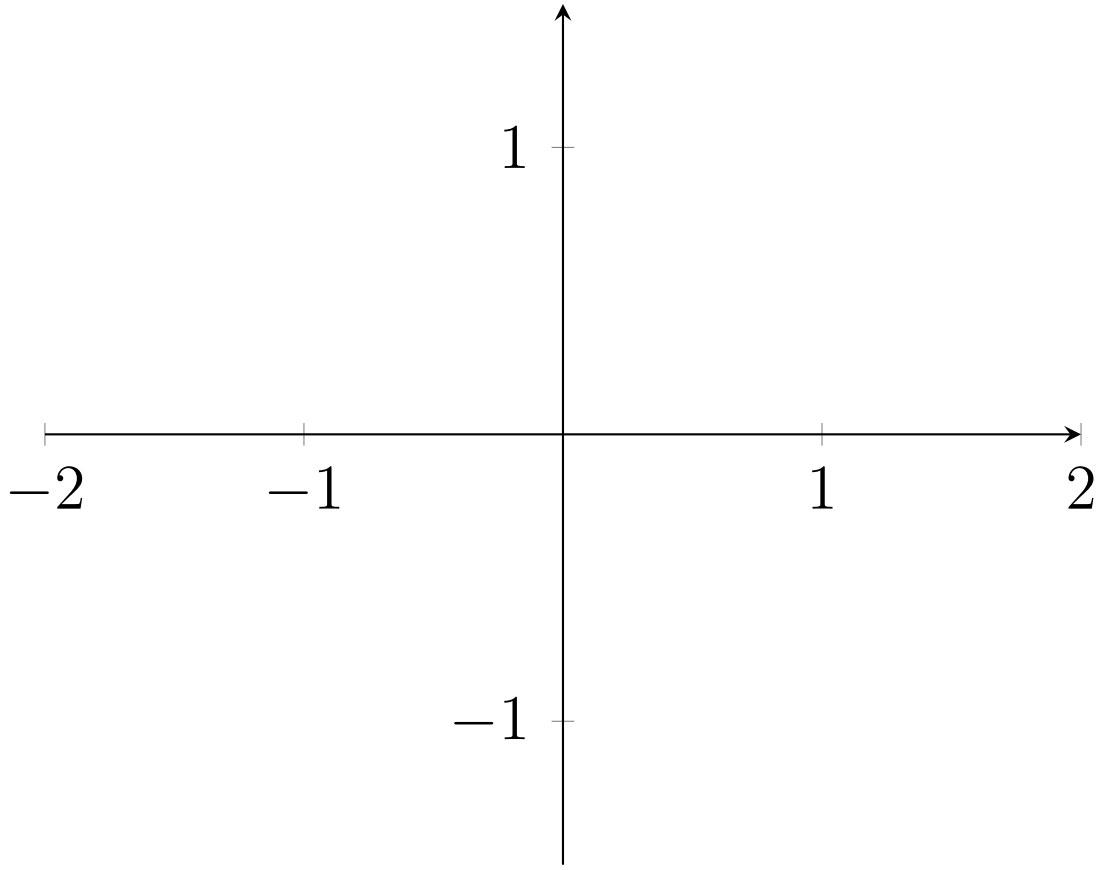
   (e) Find all three critical points of $f(x)$ and classify them as local min's, max's, or neither.

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(f) Find all inflection points of $f(x)$. Be sure to draw a number line to show each is really is an IP.

(g) Draw the graph of $f(x)$. Label all critical points and inflection points.



2. (30 points) For each of the following functions on the specified domains, find the global min and max, or explain why each one does not exist. In each case, also draw the graph of the function on the specified domain and label the points you found. Take care to draw filled or empty circles at endpoints as appropriate.

   (a) $f(x) = x^3 - x$ on $-1 \leq x \leq 2$.

   (b) $f(x) = x^4 - 2x^2$ on $-1 \leq x < 2$.

   (c) $f(x) = x^2 e^{-x}$ on $-\infty < x < \infty$.

3. (17 Points) Compute the following limits. Label any use of L'Hôpital's rule.

(a) $\lim_{x \to 0} \frac{x^2}{e^x}$

(b) $\lim_{x \to 1} \frac{\ln x}{e^x - e}$

(c) $\lim_{x \to \infty} \frac{x^{20}}{(\ln x)^{30}}$

4. (22 points) You wish to build a rectangular box with a square base and open top.

   (a) Show that if the cost of material for the base is twice the cost of material for the sides, building a cube minimizes cost. Be sure to draw and label a picture first!

   (Hint: This is quite similar to Q1 and Q4 from the optimization worksheet. Let the volume be V cubic inches and the cost of the sides be S cents per square inch. Then the cost per square inch of the base is...)

   (b) Is it always the case that building a cube minimizes cost, regardless of the ratio of the cost of base to the cost of the sides? (Hint: no need to compute further here – think through the logic and give a short worded argument.)

5. (9 points) A large cylinder is being stretched. Suppose that the volume of the cylinder is $1,000\pi$ in^3. When its radius is 10in, the radius is decreasing at a rate of 1 in/hr, while its height is increasing at 2 in/hr. Find the rate at which the surface area of the cylinder is changing at this point in time.

Challenge problem: The volume of the cylinder is always constant. By differentiating $V = \pi r^2 h$, show that it is always true that $r\frac{dh}{dt} = -2h\frac{dr}{dt}$. Use that to show that the rate of change of the surface area is $\frac{dS}{dt} = 2\pi(2r - h)\frac{dr}{dt}$.

6. (8 points) Each of the following statements is *false in general* (though may be true for particular instances). For each, briefly explain why it is false in general.

   (a) If $f(x) > g(x)$ for all $x > 0$, then $f(x)$ dominates $g(x)$.

   (b) If a is in the domain of $f(x)$ and $f''(a) > 0$, then $f(x)$ has a local minimum at $x = a$.

   (c) If a is in the domain of $f(x)$ and $f''(a) = 0$, then $f(x)$ has an IP at $x = a$.

   (d) If a is in the domain of $f(x)$ and $f'(a) = 0$, then $f(x)$ has a local max or a local min at $x = 0$.

Bonus Question (2 points) Tell me one thing (or more) you've liked about the format we've had in 105L and one thing (or more) you'd like to see changed.

Bonus Question (2 points) Draw a funny picture, paste in a good meme, tell me a good story, or tell me a good joke.